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Tunnelling undercrossing existing live MRT tunnels ☆

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ABSTRACT

In conjunction with the government of Singapore's vision to have efficient and extensive Mass Rapid Transit (MRT) system across Singapore Island, more and more rail tunnels are planned to be constructed in close proximity to sensitive buildings and infrastructures in recent years. This being a particular challenge for major infrastructure works in the urban environment. In contract C937 of Downtown Line Stage 3 (DTL3), twin bored tunnels connect Fort Canning station to Bencoolen Station, under Fort Canning Hill, with the alignment overcrossing the existing North East Line (NEL) rail tunnel, undercrossing North South Line (NSL) and Circle Line (CCL) with clear distance of less than one bored tunnel diameter. This inevitably poses great challenges to tunnelling over and under existing live tunnels without disruption to daily operation of MRT. Further compounding the challenge is the ground condition comprising of Jurong Formation steeply variable sedimentary deposits transitioning into Fort Canning Boulder Bed a colluvial deposit of quartzite boulders in a stiff clay matrix. To successfully negotiate this challenge, requires all involved parties to make contributions in their areas of expertise and work as team together to ensure that the works progress smoothly and as planned. The designer provides an overall approach demonstrating that the work can be safely carried out and provides key parameters for the contractor who carries out the tunnelling with construction control, which in turn is supervised by the supervision engineers ensuring work is carried out safely. The client, who provides the vision in the first place, has overall responsibility for the work providing continuity and ensuring that the component parts of the job are co-ordinated and carried out successfully. In addition, different monitoring levels were also established to reduce the train speed if necessary and also bus bridging service as a contingency measure. This paper summarizes the approach to the tunnelling, key tunnelling parameters during undercrossing of existing live MRT tunnels, instrumentation monitoring as well as measured railway track and structure movements due to proximity tunnelling. It reflects upon the on crucial contributory factors that ensure success within the project and serves as a reference as to how engineering boundaries can be challenged underground.

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1. Introduction

This paper focused primarily on the construction of C937 tunnels from Fort Canning Station to Bencoolen Station which are 790 m in length. The tunnels are bored in ground condition of Jurong Formation (SIII/SIV) and the Fort Canning Boulder Bed (FCBB).

The alignment of the tunnels starts off at a depth of 20 m and completed at Bencoolen Station at a depth of 45 m. The reason for this depth of 45 m is that it has to undercross North South Line (NSL) and Circle Line (CCL) at Bras Basah Road before reaching Bencoolen Station at B6 level.

Along this alignment, the 2 Tunnel Boring Machines (TBM) would have to underpass several critical structures, namely North-East Line tunnels, Fort Canning Cultural Centre, the Fort Canning Road Tunnel, the National Museum Singapore, the North-South Line tunnels and the Circle Line tunnels.

Being in the heart of the city centre, the team has to work with site constraints of a small shaft for 2 TBM operations (Bukit Panjang Bound and Expo Bound).

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The remaining portions of the paper shall elaborate the various technical challenges and stakeholder engagement.

1.1. Public relations and technical challenges

To ensure that the technical challenges are timely managed, stakeholders are proactively engaged. By keeping the stakeholders informed of the status of works and safety measures in place, stakeholders are assured that the impact, if any, is brought down to acceptable minimum.

This collaboration with the stakeholders is extremely important in maintaining a vigilant posture during the critical period of undercrossing. During the intense visual inspections, the close coordination for security clearances to carry out such activity became extremely critical.

The following will elaborate how the respective stakeholders are engaged to ensure smooth completion of the Downtown Line 3 East tunnels.

1.2. Geological conditions of C937 East tunnels

East tunnel beneath Fort Canning Park from C937 Station Box connecting to C936 Station Box is mainly in an area of Jurong formation and Fort Canning Boulder Bed formation (FCBB). The Jurong Formation exists at Fort Canning Hill area below the Fort Canning Boulder Bed, with depths of between 10 m and 60 m below existing ground level. FCBB is interfaced with the underlying Jurong Formation with depths ranging from 0.5 m to 20 m from the ground surface as shown in Figs. 1 and 2 and Table 1.

Along the East tunnel alignment, Jurong Formation exists in various states from highly fractured rocks to slightly weathered rocks. The Jurong Formation of East tunnel can be described as reddish-brown and purple siltstone with layers of dark grey mudstones and greenish-grey sandstone layers. The rocks of the Jurong Formation exhibit a wide range of strength in the Fresh State, and have weathered in different ways.

Fort Canning Boulder Bed (FCBB) is a colluvial deposit comprising strong to very strong sandstone and quartzite boulders and cobbles within a very stiff soil/clay matrix. The matrix is generally described as a hard sandy clayey silt or sandy silty clay. The colour of the matrix where encountered is typically mottled, with the characteristic deep red colour in addition to red and white, yellow, brown and red and white and occasionally purple. The cobbles and boulders in the FCBB are typically moderately strong to strong light grey, yellow and brown fine to coarse grained sandstone.

Figs. 3 and 4 show actual ground condition of Jurong formation and FCBB which was observed during cutterhead intervention of East Tunnel.

The groundwater level along the tunnel alignment fluctuates from 1.0 m to 30.0 m due to mountainous terrain at the Fort Canning Hill area.

Based on geology and ground condition, TBM operation for East tunnel was carried out in closed mode with the face pressure of 2 bar in Jurong Formation and 1.5 bar in FCBB.

1.3. Cutterhead intervention before crossing NEL, NSL and CCL tunnels

Along the East tunnel alignment, there are three existing MRT lines, North-East Line (NEL), North-South Line (NSL) and Circle Line (CCL). East tunnel is located approximately 1.3 m above NEL Tunnel, 8.7 m below NSL Tunnel and 3.3 m below CCL tunnel.

In order to cross these three MRT lines without any major problem, Cutterhead intervention (CHI) was carried out to check TBM machines and replace any wear or tear cutting tools before crossing NEL, NSL and CCL tunnels. The location for CHI was chosen at least 20 m away from existing MRT lines to limit any settlement impact.

As shown in Figs. 1 and 2, NEL tunnels are in an area of Jurong formation and NSL/CCL tunnels are in an area of FCBB. Based on geological condition and previous TBM operation data, optimum

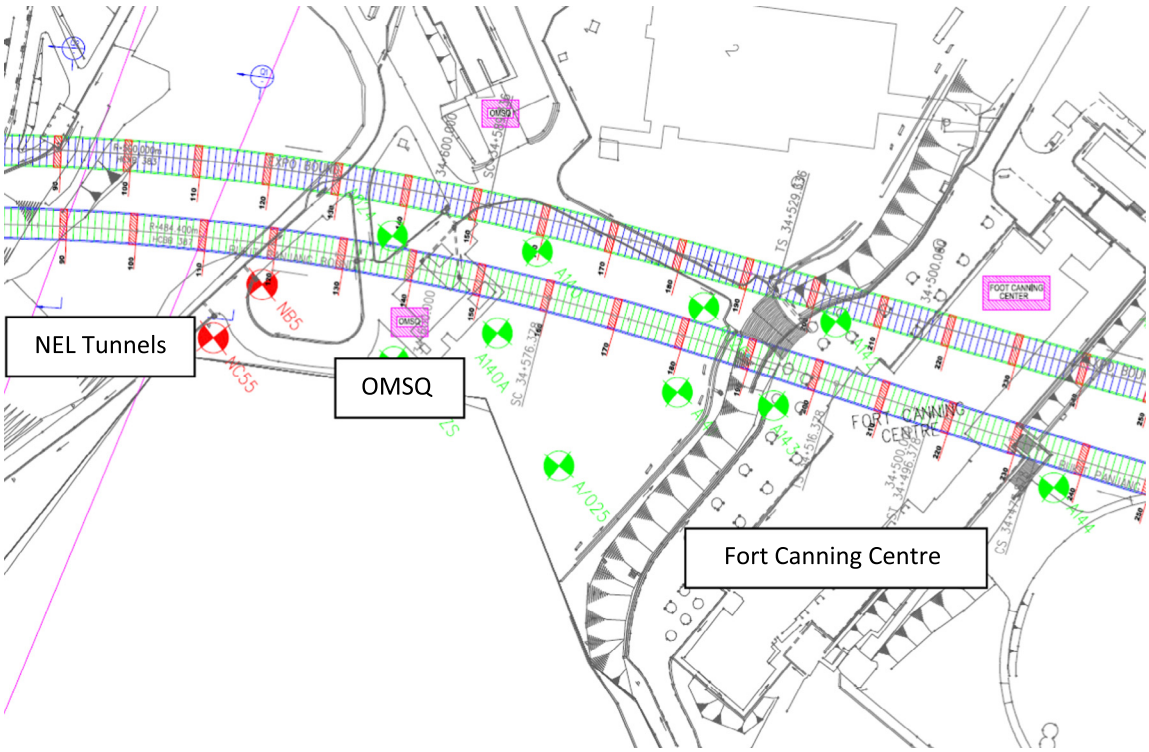


Fig. 1. Overall layout of East tunnels.

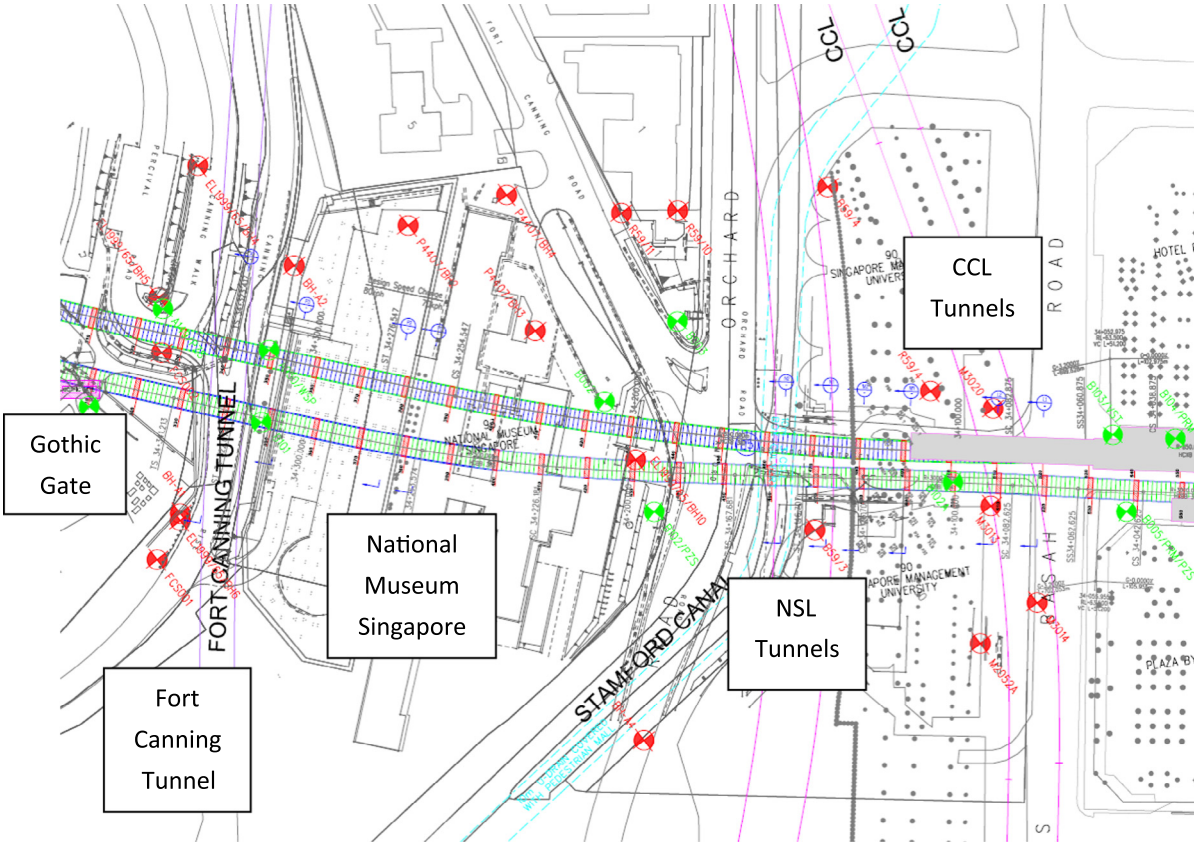


Fig. 2. Overall layout of East tunnels.

Table 1
Geological profile summary of East tunnel.

Soil type	Unconfined compressive strength	Length (m)	
		BP Bound	Expo Bound
Full Face Soil – Kallang, S(IV), S(V), S(VI)	Less than 10 MPa	95 (12%)	35 (4%)
Mixed Face – Soil & S(II)/S(III)	10 to 15 MPa	105 (13%)	105 (13%)
Full Face Jurong – S(II)/S(III)	Average 15 MPa	270 (34%)	330 (42%)
Fort Canning Boulder Bed	Average 60 MPa	325 (41%)	320 (41%)

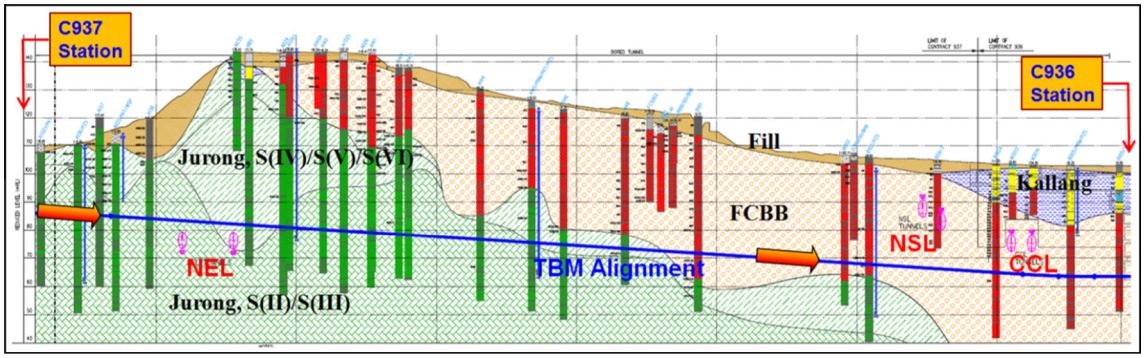


Fig. 3. Geological profile along Bukit Panjang Bound.

cutter configuration for crossing existing MRT tunnels was reviewed thoroughly.

Before crossing NEL tunnels, knife cutters were mainly used for the TBM drive. However, knife cutters were changed to roller disk cutters near NEL South Bound Tunnel due to abnormal ground condition such as high quartz content (i.e. 73.9%) and high strength (i.e. 81–109 MPa).

Fig. 5 shows cutter configuration for Bukit Panjang Bound Drive before crossing NSL tunnels. Roller disc cutters were mainly used to prevent severe cutter wear and some knife cutters were used

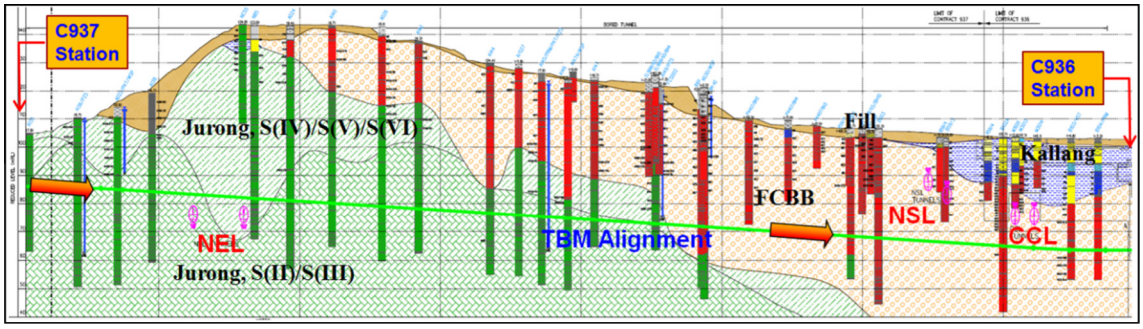


Fig. 4. Geological profile along Expo Bound.



Expo Bound, Ring P124 **Expo Bound, Ring P145** **Expo Bound, Ring P259**

Fig. 5. Actual ground condition of Jurong formation.

Table 2
Comparison of cutter wear during NSL/CCL crossing.

Tunnel drive	Before NSL crossing	Before CCL crossing	After CCL crossing
BP Bound	– CHI at P430 – Cutter configuration • #1–4: Knife cutter • #5–42: Roller cutter	– CHI at P483 – Cutter configuration • #1–42: Roller cutter – Avg. cutter wear: 2–6 mm – Cutter replacement • Single cutter: 6 Nos. (3: damaged) • Double cutter: 3 Nos. (1: damaged, 2: knife → roller)	– CHI at P537 – Cutter configuration • #1–42: Roller cutter – Avg. cutter wear: 2–6 mm – Cutter replacement • Single cutter: 13 Nos. (13: damaged) • Double cutter: 1 Nos. (1: damaged)
Expo Bound	– CHI at P420 – Cutter configuration • #1–42: Roller cutter	– CHI at P484 – Cutter configuration • #1–42: Roller cutter – Avg. cutter wear: 2–6 mm – Cutter replacement • Single cutter: 7 Nos. (2: damaged) • Double cutter: 2 Nos. (2: damaged)	– CHI at P548 – Cutter configuration • #1–42: Roller cutter – Avg. cutter wear: 2–6 mm – Cutter replacement • Single cutter: 19 Nos. (15: damaged)

at cutterhead centre part to prevent clogging which can be caused by stiff clay. Later, centre knife cutters were also replaced to roller disc cutters for CCL crossing in Bukit Panjang Bound and NSL/CCL crossing in Expo Bound as clogging was properly controlled by soil conditioning during excavation.

Table 2 shows the comparison of cutter wear between Bukit Panjang Bound drive and Expo Bound Drive during crossing NSL and CCL. Main reason to replace cutters after crossing NSL/CCL lines was not due to primary wear, loss of material of the cutting blade, but due to secondary wear such as broken ring and ring crack caused by strong boulders as shown in Fig. 6.

2. Stakeholders

2.1. North-East Line MRT tunnels (SBST CORP)

Based on the East tunnels alignment, under Fort Canning Hill, the TBMs over-crossed the North-East Line (NEL) tunnels with the tunnel invert extrados with a mere 1 m above the crown of NEL tunnels. This close proximity crossing above live line which carries hundreds of thousands of commuters daily is definitely a high severity challenge faced in the tunnelling industry of Singapore (see Figs. 7 and 8).



Fig. 6. Actual ground condition of Fort Canning Boulder Bed (FCBB).

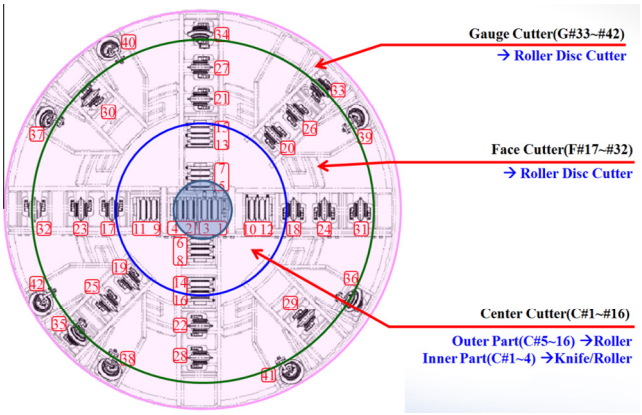


Fig. 7. Cutter configuration for the crossing of NSL and CCL.

The stakeholder for this overcrossing is SBST Corporation. Extensive close coordination was carried out. First, the installation of instruments and other works which requires track access need to be accompanied by a Track Protection Officer (TPO). These works were to be carried out during the non-revenue hours.

Together with the joint approval of Development of Building Control Rail (DBC Rail) and SBST, the extent and location of various type of instruments were established and installed (see Figs. 9 and 10).

In addition, the track instrumentation was in place to ensure that the serviceability of the track is still in good working condition (see Fig. 11).

When the TBM has reached the NEL tunnels, track survey will again be done during engineering service hours to ensure that the tracks are still in good serviceable condition for the trains to continue without the need to reduce speed. 10 days after the TBM has passed the NEL tunnels, another track survey will be conducted as part of the post-condition survey.

Other than the track gauging survey, daily visual inspections were also carried out during engineering hours, to ensure structural soundness and instrumentation maintenance like cleaning of prisms and checking of instruments data loggers.

As the train services are live, real time monitoring of the instruments becomes very important as no human access is possible, thus based on the EL beam readings, the track bed movement was closely monitored. Other than the track monitoring, the structural aspect of tunnel is also monitored by real time prism.

To ensure the safety of the commuters, several tiers of limits are set. For example, the train speed will be reduced to 40 km/h if any of the monitoring instruments shows that it breeches the predetermine level. It will be further reduce to 18 km/h or even requested for train services to be suspended if the monitoring results continue to show unfavourable trend and breeches the pre-determine level.

Please see *below table* for the various levels of the monitoring parameters.

During engineering service hours, the instruments are check and reconcile with manual survey or track gauge survey (see Table 3).

The DTL3 tunnels crossing over NEL was successful completed on 27 September 2013 for BP bound and 19 January 2014 for Expo Bound.



Fig. 8. Wear of roller disc cutters in FCBB.

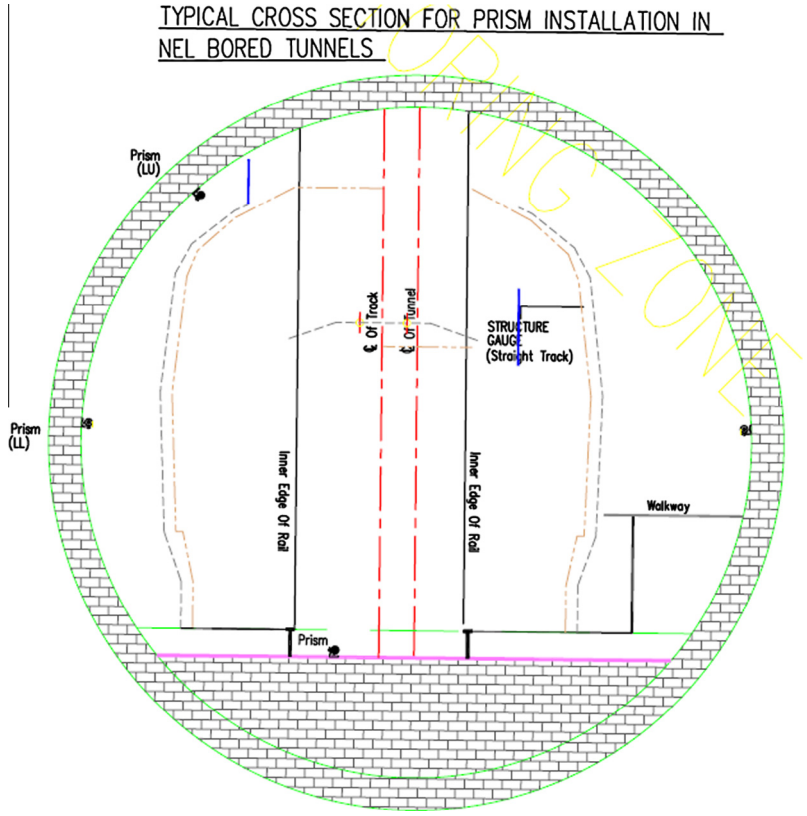


Fig. 9. Instrumentation layout for prisms installed in the NEL tunnels.

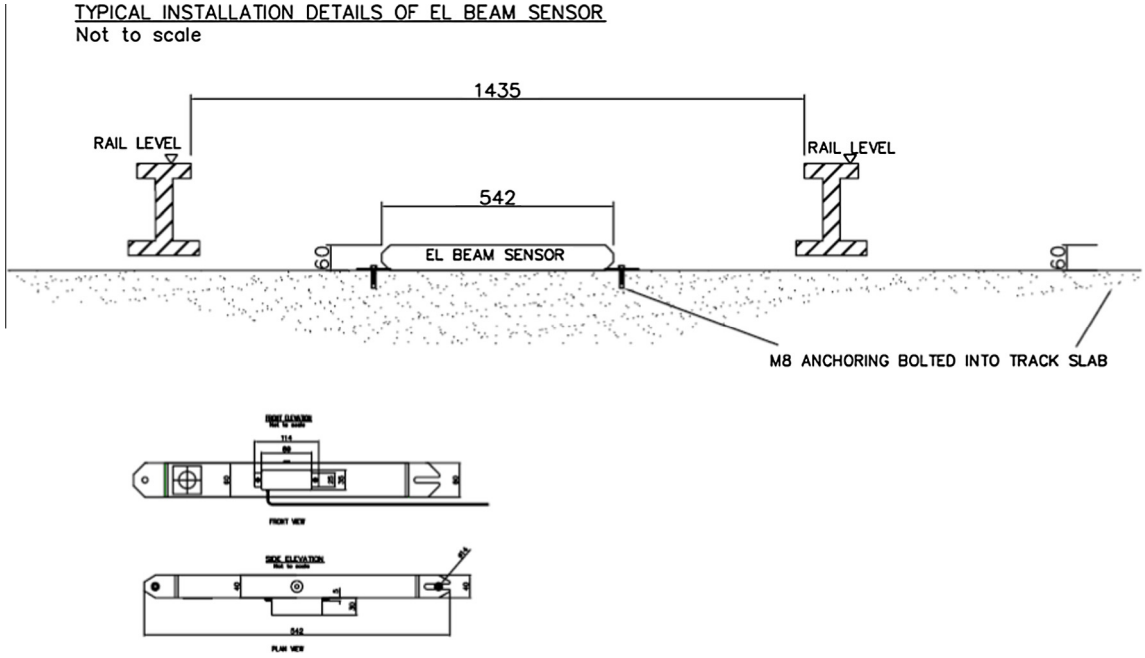


Fig. 10. Installation layout of electro-level beams for measuring track movements.

2.2. North-South Line and circle line tunnels (SMRT Corp)

The North South Line (NSL) and the Circle Line (CCL) tunnels are also of high challenge and concern as the undercrossing took place at 8 m below NSL while for CCL, it is nearer with only a vertical distance of 3.3 m. Again, the NSL and CCL are live tunnels with hundreds of thousands of commuters daily. Any delay or worse, suspension of train services will cause mega disruption to the transport system in the city area. Thus close coordination with DBC and the Operator SMRT to ensure train serviceability during the construction or if the pre-determine monitoring levels are breeched, contingency measures such as bus bridging services will

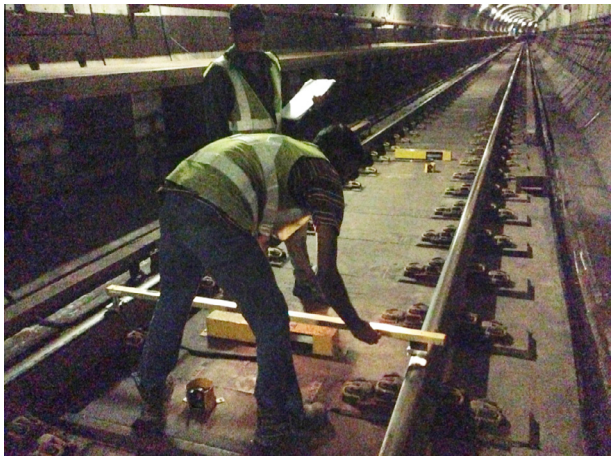


Fig. 11. Photograph showing track instrumentation recording for track gauge and twist.

be activated to alleviate the transport system. At all time, the safety of the commuters are placed at highest priority and the instructions are clear, Stop the train services as necessary and reconfirmed with manual inspections and survey, rather than to have any train derailment which will cause injuries to commuters.

Likewise as with NEL (SBST), daily visual checks and instrumentation maintenance were carried out when the TBMs are within the influence zone of NSL and CCL.

The instruments were carried out at least half a year before the actual crossing so as to establish the baseline readings.

Procedures for track access application, the approvals and the safety to take note were also properly coordinated and agreed in the various coordination between the Project team and SMRT. With the finalised agreement of all parties, SMRT issued a circular within the organisation to regularise the agreement and established the authorised point of contact for the purpose of this construction and the safety to take note of (see Figs. 12 and 13).

3. Performance of tunnelling under NEL

The NEL crossing is the first crossing of live MRT tunnels and hence it was flagged with utmost concern and attention. Nonetheless, the undercrossing was not without challenges.

North bound crossing was smoother drive as compared to Sound bound and the tunnelling parameter in general as follow (see Tables 4 and 5).

3.1. Challenge in conditioning the sandstones

Due to steep variable ground strata in Jurong Formation, the construction team initially faced difficulty to achieve the right mix of soil conditioning, in particularly a sudden and large degree of sandstone was encountered during first crossing of NEL tunnels. As time goes by, larger rock fraction was gradually sunk to the bottom of excavation chamber and piled up exerting additional force restraining the screw conveyor from rotating. The condition worsened when large amount of fine was being mucked out leaving

Table 3
Review parameters for tunnelling works above NEL.

Emergency Response for any movement of MRT over 15mm			
1. For Track (NEL/NSL/CCL)			
Dip /Peak	Reduce Speed to	Versine	Reduce Speed to
1:250	40km/hr	1:179	40km/hr
1:125	18km/hr	1:110	18km/hr
1:62	Suspend Service and realign track	1:92	Suspend Service and realign track
Twist (over 2.5 / 3m)	Reduce Speed to	Twist (over 15.8m)	Reduce Speed to
1:500	40km/hr	1:451	40km/hr
1:416	18km/hr	1:395	18km/hr
1:263	Suspend Service and realign track	1:263	Suspend Service and realign track
2. For Structure (Tunnel)			
1. Any movement over 10mm structure Inspection to be carried out at earliest opportunity			
2. Ovalisation (Vert Change) 30mm suspend service for immediate inspection			
3. Squat (Horizontal Change) 30mm suspend service for immediate inspection			
4. Total movement in any direction 60mm suspend service			
Note :			
1)Track values are absolute and can only be proposed once we have reviewed the Track survey and the actual condition of the track.			
2) Structure values are relative (i.e. movement are baeline has been established)			
* Reference			
Allowable Limits for Movement on CPRP (2004)			
Movement (mm) Table 9.1(b)			
Total Movement in Any Direction		Differential Movment in Any Plane	
15		15 or 1:1000	
		Whichever is lesser	
Allowable Limits for Track Distortions on CPRP (2004)			
Twist / Change in Cant		Vertical DIP or Peak	Change in Horizontal Versine
1.5mm (measured over 2.5m) or 1:1666		5mm	6mm
		measured over 5m Chord	measured over 16 m Chord
Target Value for Existing Tunnel settlement			
MRT	Geology	Volume loss	Target Volume by Tunnel Drive
NEL =1.28 m below C937	SIII	0.85%	7.2mm
NSL =10.04m above C937	FCBB	0.85%	11mm
CCL = 3.28m above C937	FCBB	0.75%	13mm
Transport Authority			GS E&C

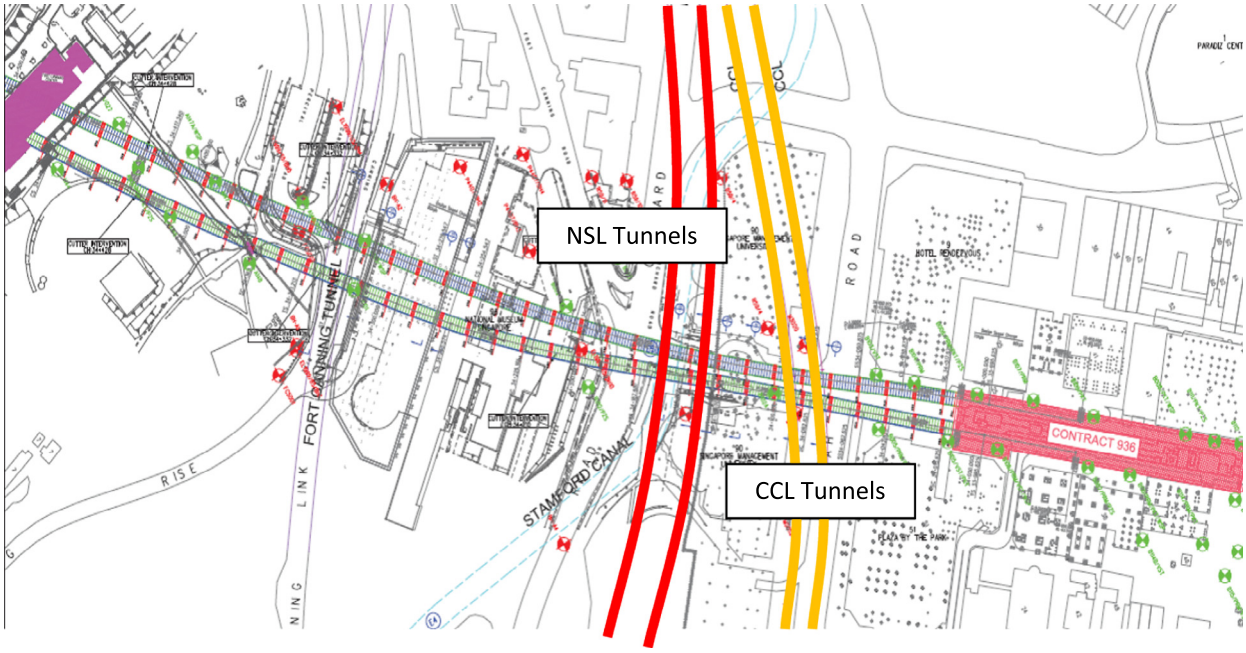


Fig. 12. Layout of tunnel alignments undercrossing NSL and CCL tunnels.

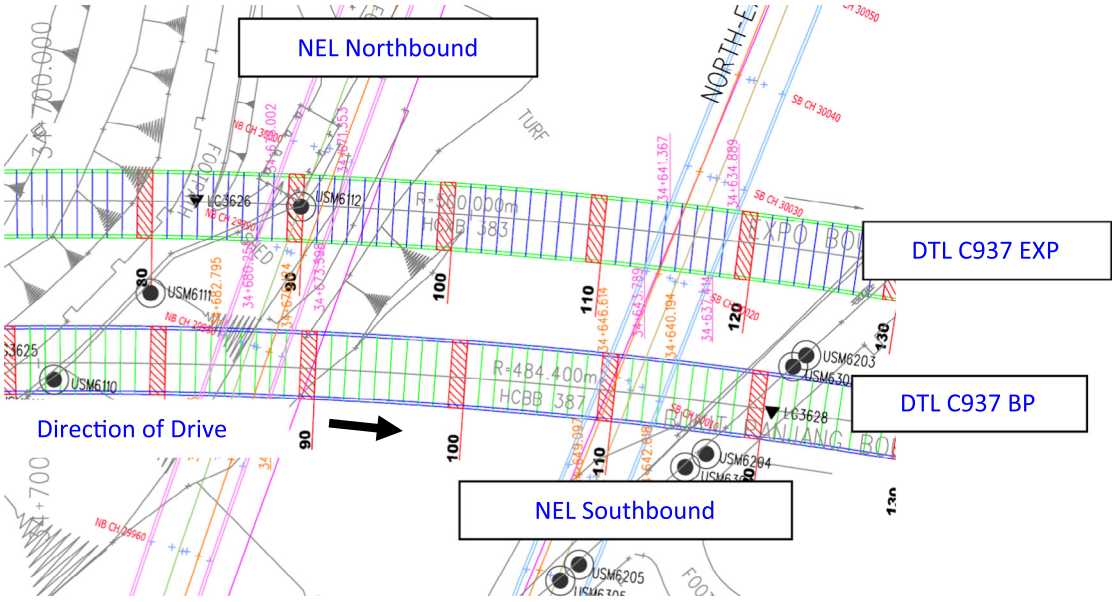


Fig. 13. Overall layout of NEL crossing.

Table 4
Summary of parameters during undercrossing of NEL North Bound tunnel.

Face pressure	~2.0 bar
Thrust force	13,000–15,000 kN
Torque	2000–2500 kN m
Advance rate	7–10 mm/min

Table 5
Summary of parameters during overcrossing of NEL South Bound tunnel.

Face pressure	~2.0 bar
Thrust force	23,000–32,000 kN
Torque	2000–2500 kN m
Advance rate	Less than 5 mm/min

mostly the rock fragments inside excavation chamber while correct soil conditioning was still trying to be achieved. Bentonite was eventually discharged into the excavation chamber and remixing was done considerably between separated soil and rock fragments so as to attain desired soil paste.

3.2. Challenge faced in highly abrasive sandstones

On the other hand, encountering of hard and highly abrasive sandstones in the way of crossing had greatly affected advance rate. The said sandstone has average Unconfined Compressive Strength (UCS) of 60 MPa which is 4 times higher than average UCS in Jurong Formation SIII/SIV. Both face and gage knife cutters have worn to an extensive degree resulted in higher thrust force

was required to impact the rock, break the frictional force and to manoeuvre the TBM in curve alignment. At a point in time, the project team was extremely concern about having to carry out cutter head intervention right above the tunnel. Fortunately, the machine was able to carry through the completion of the crossing for cutter tool change and maintenance.

3.3. Instrumentation monitoring

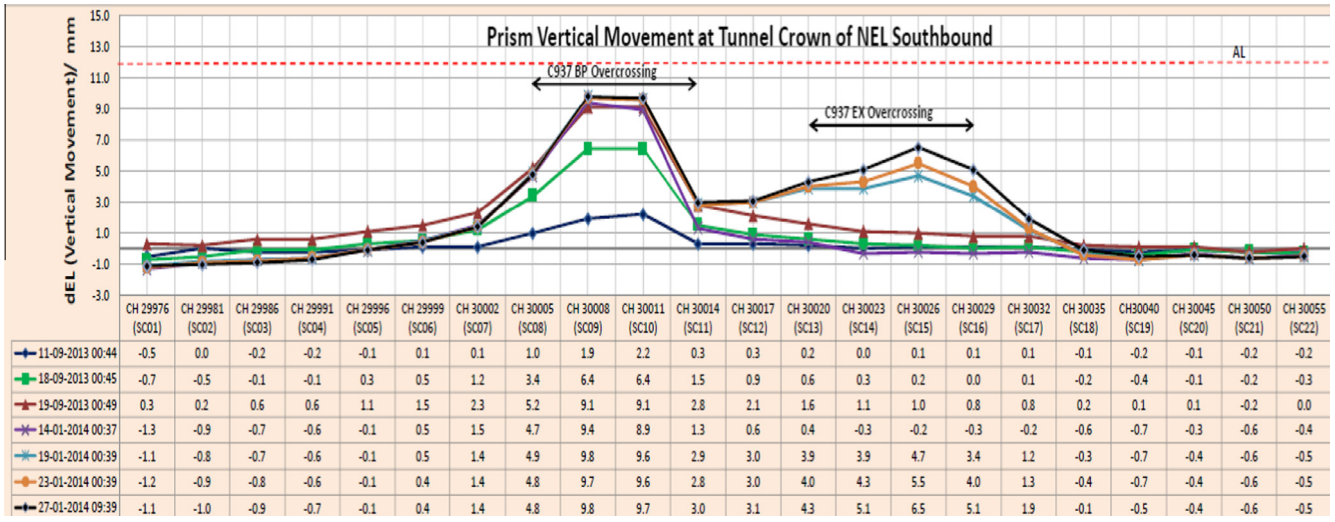
Ovalization was observed at two distinct locations of NEL tunnels where Bukit Panjang Bound and Expo Bound over-crossed. The largest vertical upwards movement was recorded by prism located at crown of NEL Southbound during the first tunnel drive which was also the location where encountering of highly abrasive sandstone occurred (see Fig. 14).

Following the completion of tunnelling works overcrossing the NEL tunnels, assessments carried out on the NEL tunnels in the aspects of structure, track and Overhead Caternary System confirmed no impact due to the tunnelling works. Subsequent coordination with SBST to formally close out the overcrossing was successfully completed.

4. Tunnelling under north south line and circle line tunnels

4.1. Overall challenges of undercrossing NSL and CCL

Information such as proportion of clay to boulder or Boulder Volume Ratio was unable to be concluded in normal site investigation which has increased the difficulty in predicting the planned cutterhead intervention location. Nevertheless, it was laid out in Geotechnical Interpretative Baseline Report (GIBR) that Sandstone boulders in FCBB has CERCHAR index ranging from 4 to 7 which is classified as extremely to Quartzitic abrasive. Besides, average Unconfined Compressive Strength (UCS) of sandstone boulders in FCBB is 60 MPa but could range up as high as 200 MPa. On the other hand, the moisture content of soil matrix is at or very close to plastic limit and can be described as silt with high to very high plasticity. This information enables contractor to foresee the geological challenges and potential risks in particular during crossing of sensitive structures. During the planning stage, the project team had identified a few locations for planned cutterhead intervention to check and maintain top-notch condition of cutter tools prior to crossing of each sensitive structure.



	CH 29976 (SC01)	CH 29981 (SC02)	CH 29986 (SC03)	CH 29991 (SC04)	CH 29996 (SC05)	CH 29999 (SC06)	CH 30002 (SC07)	CH 30005 (SC08)	CH 30008 (SC09)	CH 30011 (SC10)	CH 30014 (SC11)	CH 30017 (SC12)	CH 30020 (SC13)	CH 30023 (SC14)	CH 30026 (SC15)	CH 30029 (SC16)	CH 30032 (SC17)	CH 30035 (SC18)	CH30040 (SC19)	CH 30045 (SC20)	CH 30050 (SC21)	CH 30055 (SC22)
11-09-2013 00:44	-0.5	0.0	-0.2	-0.2	-0.1	0.1	0.1	1.0	1.9	2.2	0.3	0.3	0.2	0.0	0.1	0.1	0.1	-0.1	-0.2	-0.1	-0.2	-0.2
18-09-2013 00:45	-0.7	-0.5	-0.1	-0.1	0.3	0.5	1.2	3.4	6.4	6.4	1.5	0.9	0.6	0.3	0.2	0.0	0.1	-0.2	-0.4	-0.1	-0.2	-0.3
19-09-2013 00:49	0.3	0.2	0.6	0.6	1.1	1.5	2.3	5.2	9.1	9.1	2.8	2.1	1.6	1.1	1.0	0.8	0.8	0.2	0.1	0.1	-0.2	0.0
14-01-2014 00:37	-1.3	-0.9	-0.7	-0.6	-0.1	0.5	1.5	4.7	9.4	8.9	1.3	0.6	0.4	-0.3	-0.2	-0.3	-0.2	-0.6	-0.7	-0.3	-0.6	-0.4
19-01-2014 00:39	-1.1	-0.8	-0.7	-0.6	-0.1	0.5	1.4	4.9	9.8	9.6	2.9	3.0	3.9	3.9	4.7	3.4	1.2	-0.3	-0.7	-0.4	-0.6	-0.5
23-01-2014 00:39	-1.2	-0.9	-0.8	-0.6	-0.1	0.4	1.4	4.8	9.7	9.6	2.8	3.0	4.0	4.3	5.5	4.0	1.3	-0.4	-0.7	-0.4	-0.6	-0.5
27-01-2014 09:39	-1.1	-1.0	-0.9	-0.7	-0.1	0.4	1.4	4.8	9.8	9.7	3.0	3.1	4.3	5.1	6.5	5.1	1.9	-0.1	-0.5	-0.4	-0.6	-0.5

Fig. 14. Recorded vertical movement of NEL live tunnel during undercrossing by DTL3 drives.

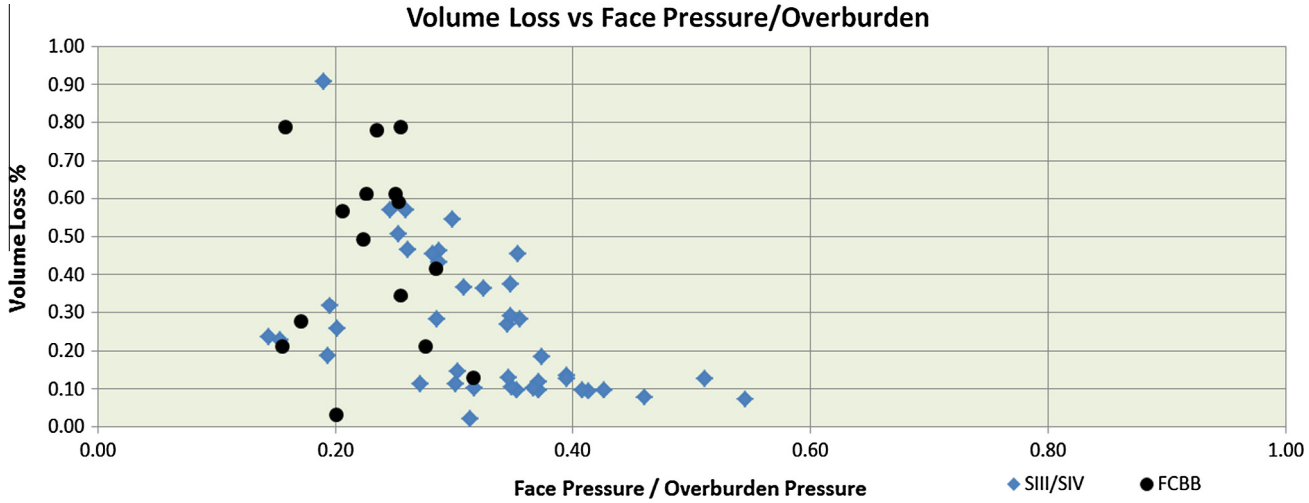


Fig. 15. Relationship between normalised face pressure and volume loss for EPB tunnelling in FCBB.

Mixed ground earth pressure balance (EPB) tunnel boring machine (TBM) with cutterhead opening ratio of approximately 32% was selected for East Tunnel Drive upon consideration of safety, geological profile as well as challenges (shallow cover to live MRT tunnels, mixed face condition, time and cost effectiveness). In addition, screw conveyor is capable of mucking out rock size up to 325 mm diameter.

4.2. Tunnelling strategies in FCBB geology

For selection of cutter tools configuration, a combination of tungsten carbide knife cutters at the centre while 17 in. wedge lock cutter discs were installed at the remaining outer positions at cutterhead. As there was not much past tunnelling experience in FCBB available, the project team initially foresaw that clay were likely to be piled up or clogged at the centre cutterhead which in return will prevent the cutter discs to rotate and subsequently wear the tools excessively, knife cutters at the centre should be of more efficient cutters in this context. Nevertheless, front knife cutters were found worn off severely by direct impact of hard boulders after first drive of Bukit Panjang Bound crossing NSL tunnel. The project team then decided to replace front knife cutters with disc cutters for the remaining drives which proved to be more effective with the correct usage of clay dispersant agent.

To keep the target volume loss of 1% in general, applied face pressure in FCBB was ranging from 0.15 to 0.30 times the total overburden pressure. Figure below shows the relationship between normalised face pressure and volume loss (computed from recorded surface settlement by empirical formula) for EPB tunnelling in FCBB as well as Jurong Formation SIII & SIV. In order to prevent undesired ground loss or overpressuring the soil, operation range of face pressure was set at target face pressure with 0.2 bar variation. During crossing of live tunnel, applied face pressure was always maintained at range closer to higher limit (see Fig. 15).

Apart from good excavation volume management, soil conditioning is essential in forming soil plug at bulkhead gate for maintaining face pressure within operational range. Besides, good soil



Fig. 16. Conditions of cutters as observed in findings during cutterhead intervention.

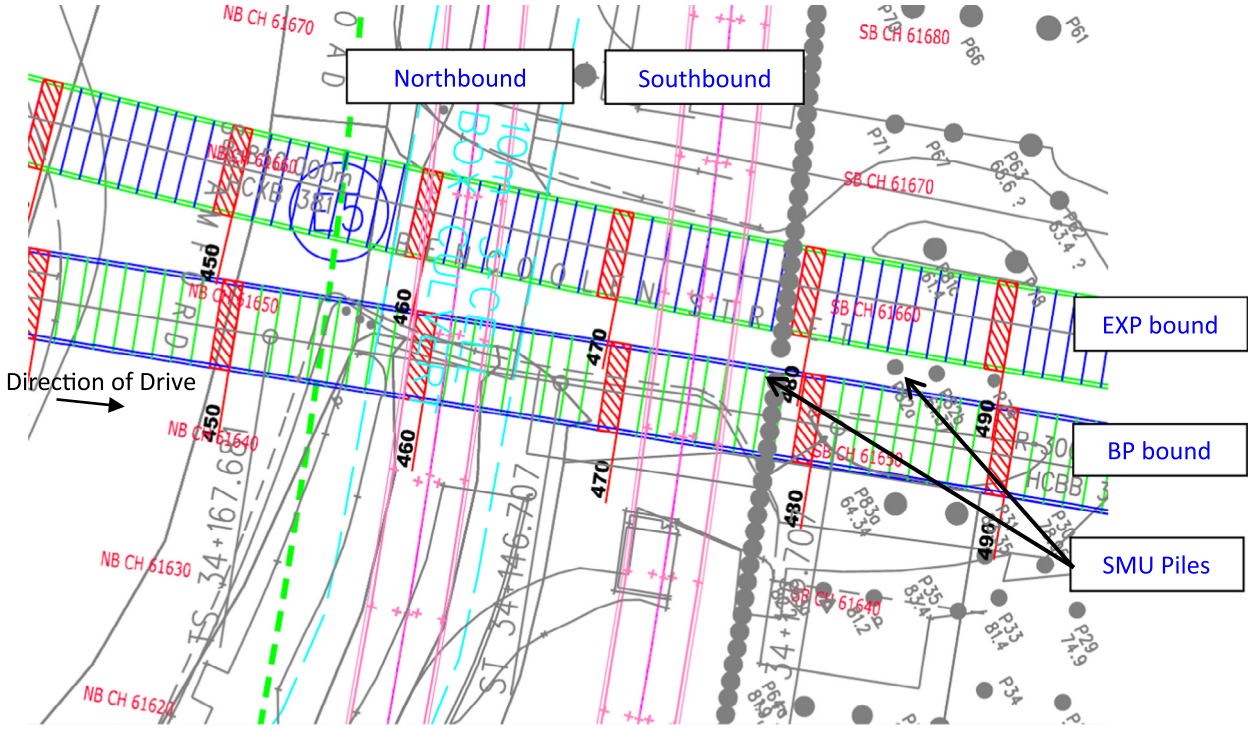


Fig. 17. Alignment layout plan of C937 tunnels and NSL tunnels.



Fig. 18. Settlement Trendplot for second drive TBM.

Table 6
Summary of parameters during undercrossing of NSL tunnels.

Face pressure	~1.7 bar
Thrust force	16,000–20,000 kN
Torque	2000–2400 kN m
Advance rate	10–13 mm/min

conditioning also eased the mucking-out of boulders fraction, reduced cutterhead torques and the wearing rate of cutters while prolonging its serviceability period. So far, there was no jamming of screw auger for the two tunnel drives where boulders were broken into fractions that could fit well within the pitch of screw conveyor. To enable the disc cutters to roll across boulders, soil conditioning with anti-clogging additives was deployed to disperse the clay from lodgement of disc cutters.

In order to minimize accumulation of compressed air and to ensure excavation chamber was always full of muck, an air release value which was installed behind the excavation chamber and located close to the crown was periodically opened so as to evacuate air built-up by excessive use of foam or muck discharge. On top of the aforesaid measure, the "apparent density" of muck in excavation chamber, which is computed based the two registered face pressures at different height, was also used in detecting the possibilities of air accumulation.

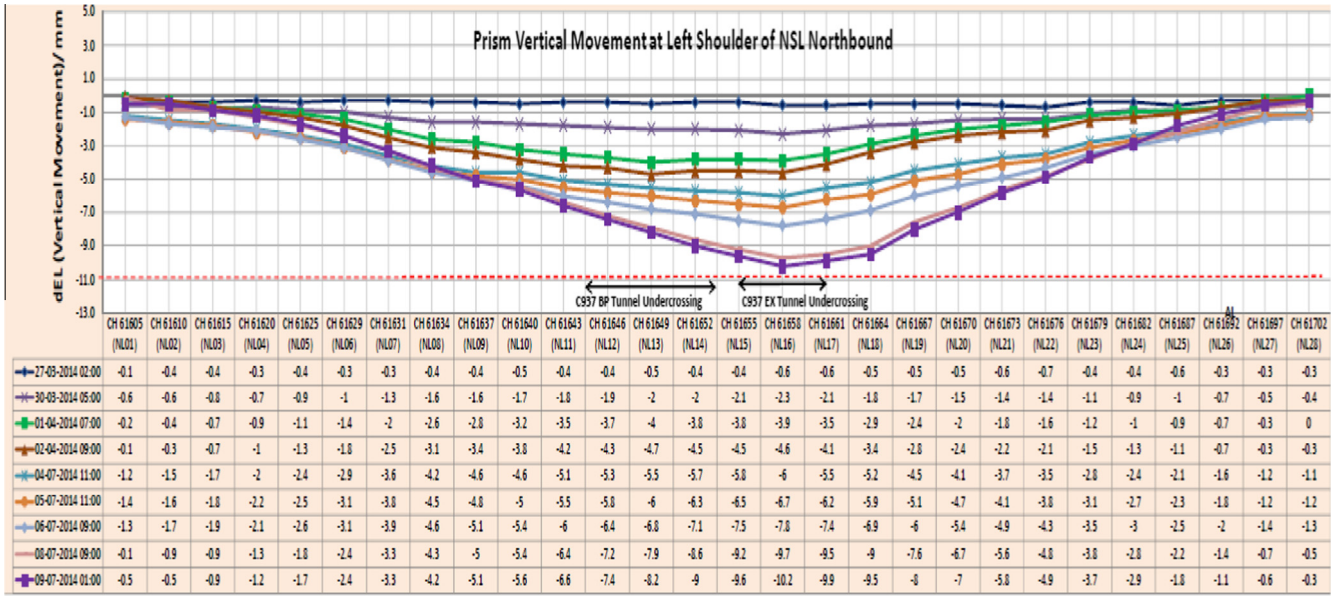
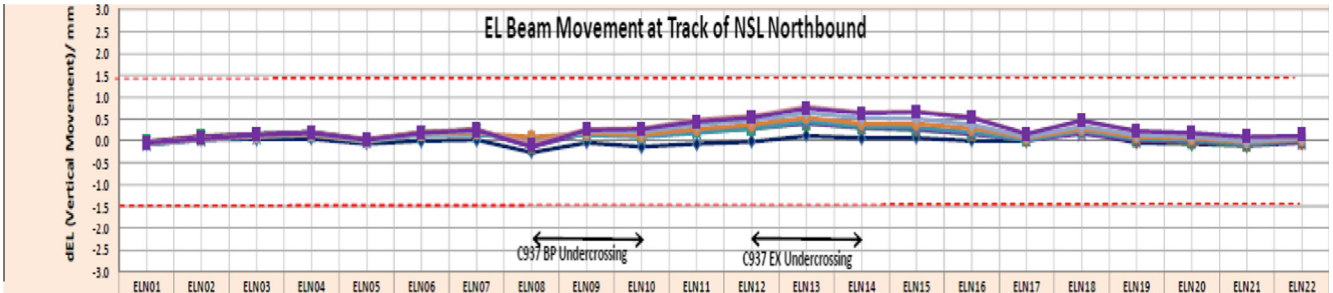


Fig. 19. Recorded vertical movement of NEL live tunnel during undercrossing by DTL3 drives.



	ELN01	ELN02	ELN03	ELN04	ELN05	ELN06	ELN07	ELN08	ELN09	ELN10	ELN11	ELN12	ELN13	ELN14	ELN15	ELN16	ELN17	ELN18	ELN19	ELN20	ELN21	ELN22
27-03-2014 13:02	-0.0920	0.0231	0.0323	0.0457	-0.0613	-0.0162	0.0132	-0.2601	-0.0401	-0.1452	-0.0770	-0.0255	0.1088	0.0556	0.0709	0.0005	-0.0137	0.1972	-0.0490	-0.0752	-0.1226	-0.0522
31-03-2014 02:02	-0.0307	0.0859	0.1478	0.1898	0.0048	0.1857	0.2340	-0.1431	0.1694	0.0938	0.1946	0.2607	0.3822	0.2839	0.2453	0.1461	0.0202	0.2171	0.0080	-0.0300	-0.1037	-0.0156
01-04-2014 12:02	-0.0266	0.0939	0.1569	0.2015	0.0278	0.2026	0.2550	-0.1313	0.2041	0.1656	0.2500	0.3209	0.4307	0.3096	0.2723	0.1647	0.0273	0.2084	0.0065	-0.0267	-0.0791	-0.0352
02-04-2014 10:03	-0.0306	0.0805	0.1440	0.1820	0.0143	0.1769	0.2232	-0.1483	0.1726	0.1238	0.2108	0.2850	0.3953	0.2935	0.2571	0.1538	0.0242	0.1387	0.0068	-0.0165	-0.0717	-0.0344
04-07-2014 12:02	-0.0871	0.0249	0.0755	0.1241	-0.0197	0.1219	0.1371	0.0547	0.1191	0.0747	0.1811	0.2741	0.4256	0.3152	0.3004	0.1961	0.0253	0.2120	0.0413	0.0157	-0.0630	0.0143
05-07-2014 14:03	-0.0694	0.0310	0.0941	0.1367	-0.0066	0.1432	0.1759	0.0777	0.1663	0.1218	0.2643	0.3585	0.5252	0.4010	0.3903	0.2731	0.0631	0.2675	0.0866	0.0553	-0.0271	0.0360
06-07-2014 09:30	-0.0647	0.0319	0.1011	0.1444	-0.0114	0.1580	0.2049	-0.0694	0.2066	0.1866	0.3611	0.4618	0.6420	0.5202	0.5068	0.3668	0.0917	0.3195	0.1268	0.0957	-0.0073	0.0556
08-07-2014 09:30	-0.0596	0.0374	0.1139	0.1596	0.0238	0.1870	0.2394	-0.0659	0.2481	0.2769	0.4699	0.5700	0.7832	0.6470	0.6738	0.5402	0.1462	0.4550	0.2064	0.1777	0.0865	0.1045
09-07-2014 01:30	-0.0591	0.0400	0.1174	0.1604	0.0133	0.1789	0.2318	-0.1333	0.2416	0.2487	0.4350	0.5314	0.7332	0.6321	0.6498	0.5248	0.1420	0.4528	0.2123	0.1790	0.0835	0.1100

Fig. 20. Recorded EL beam movement of CCL track during undercrossing by DTL3 drives.

To achieve the pre-determined volume loss of 0.75%, annulus grouting is paramount important to be filled soonest possible so as to minimize the gap closure between overcutting profile and TBM shield. Nevertheless, cutterhead rotation was set to be operated at low revolution for the ease of detection of torque spike as an indication of obstruction encounter during the course of crossing sensitive structures. To maintain the desire advance speed, larger thrust force was applied to increase the depth of cutting of cutter tools. Attention was required to monitor lateral movement of tunnel in ensuring that the drag force generated has insignificant impact to tunnels in close proximity. In addition, cutterhead contact force was also monitored so as to detect early sign of blunt cutters and as part of the prevention measures for propagation of broken ring from one roller cutter to adjacent cutters. As highlighted in above section "Cutterhead Intervention before Crossing NEL, NSL and CCL tunnels", large numbers of disc cutters were found to be failed predominantly in bearing failure of roller ring. This premature failure is likely caused when roller ring is cutting from relatively soft clay to hard boulder where concentrated area of stress occurred at two different depth of cutting (see Fig. 16).

Tail void grouting or commonly known as backfill grouting is critical operation in controlling ground movement. During excavation, approximately 120% of theoretical volume of backfill grout was continuously pumped via 4 ports up to 1 bar higher than the target face pressure. Vigorous maintenance of backfilling grout ports was done regularly in particular during ring building period with the aim to reduce downtime during excavation. The quality of hardened grout was then re-assessed by probe drilling method at locations closer to crown in every alternate rings.

5. Performance of tunnelling under NSL

The geology condition at undercrossing of NSL tunnels is FCBB with both tunnels situated at approximately 2 tunnel diameter apart but at a different level of 5.6 m. DTL 3 tunnel drives will first pass through NSL Northbound and follow with NSL Southbound. Moving forward 6 m from Southbound, a row of existing contiguous bored piles (CBP) at Singapore Management University (SMU) was anticipated to be encountered at the crown of TBM. The project team planned and prepared for the likelihood scenario where unplanned cutterhead intervention has to be carried out within the First MRT Reserve Line in the event of CBP hindering the advance of TBM drive (see Fig. 17).

Fortunately, both DTL3 tunnel drives passed through CBP location smoothly without encountering of any obstruction. No abnormal spike in cutterhead torque nor sign of rebar or concrete were found at discharged muck. Besides, both tunnel drives had also successfully manoeuvred through 500 mm tight clearance between SMU linkway piles & TBM. In early design stage, impact assessment on induced bending moment, shear as well as reduction in geotechnical capacity of existing piles has been conducted (see Fig. 18).

The general tunnelling parameters by DTL3 Drive of Bukit Panjang Bound and Expo Bounds Tunnel are as follows (see Table 6).

Although NSL Southbound tunnel is located closer to both tunnel drives, the largest recorded vertical movement was at NSL Northbound tunnel. Vertical downward movement of 10.5 mm, which is below the predicted movement, was registered once TBM tail of Expo Bound drive (second drive) left NSL Northbound tunnel. Nevertheless, this value eventually crept to 11.7 mm (see Fig. 19).

Despite that recorded vertical movement of tunnel structures has slightly exceeded the predicted value, the reading of electro level (EL) beams remain healthy throughout the entire tunnel drives. Maximum Dip of 0.7 mm is registered but the corresponded Twist is well below the alert level of 1:2350 (see Fig. 20).

Apart from monitoring of existing tunnel, the newly completed tunnel of DTL3 Panjang bound was also being monitored by convergence prisms during close proximity tunnelling by Expo Bound drive. Squat was observed at the newly completed tunnel within the close proximity tunnelling zone but the magnitude recorded is only 4 mm (see Fig. 21).

6. Performance of tunnelling under CCL

At the location of crossing, both the existing Outerbound and Innerbound of CCL structures are located at reduced level of approximately 100 mm different. Likewise, the proposed vertical alignment of both DTL 3 Bukit Panjang Bound and Expo Bound at crossing is at almost similar top of rail level. On the other hand, the design line for both Bukit Panjang Bound Expo Bound is such that the closest distance between extrados is merely 2 meter apart so as to avoid existing piles of Singapore Management University.

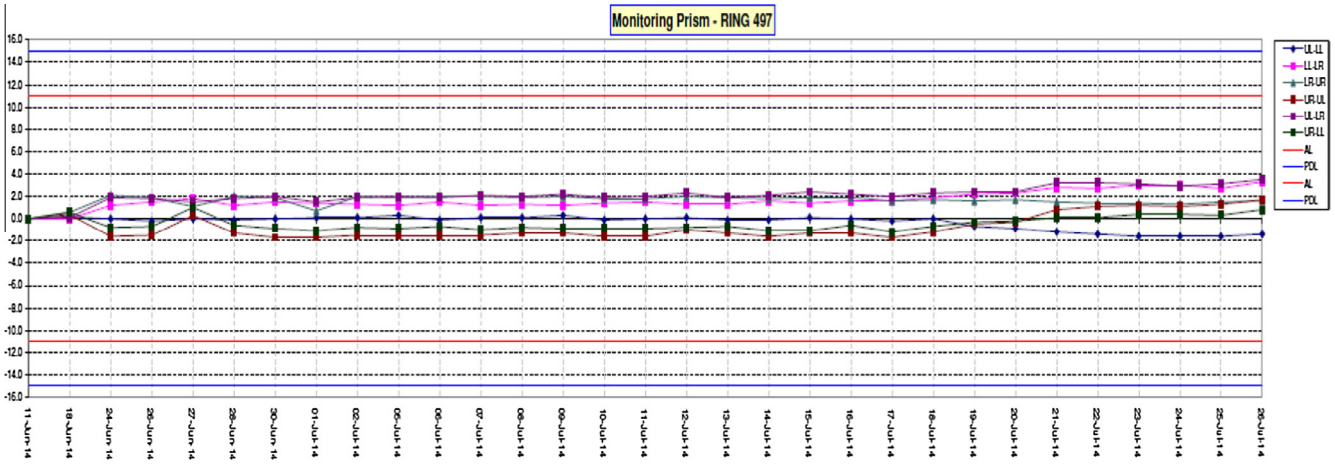


Fig. 21. Recorded prism movement of Bukit Panjang tunnel during Expo Bound drives.

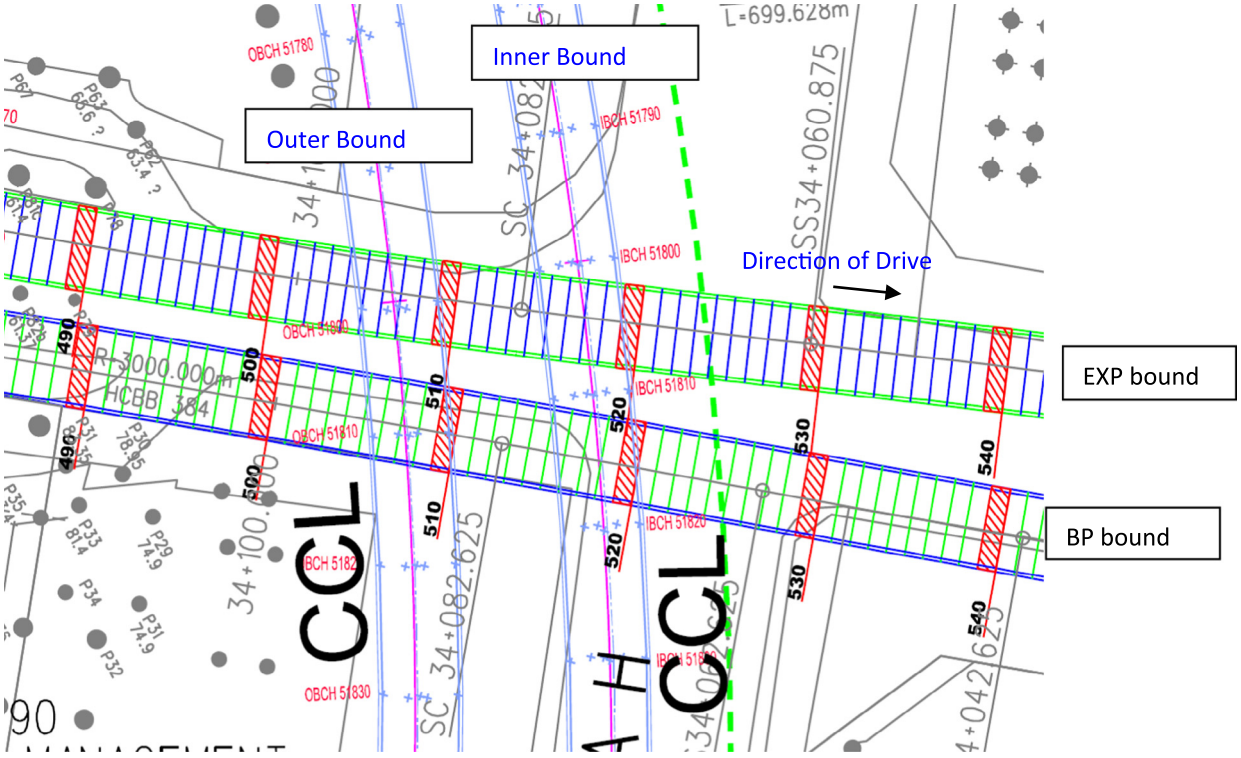


Fig. 22. Alignment layout plan of C937 tunnels and CCL tunnels.

Table 7
Summary of parameters during undercrossing of CCL tunnels.

Drive	Bukit Panjang Bound	Expo Bound
Face pressure	~1.7 bar	~1.7 bar
Thrust force	16,000–28,000 kN	13,000–15,000 kN
Torque	1600–3000 kN m	1600–2300 kN m
Advance rate	8–11 mm/min	10–15 mm/min

The undercrossing of CCL tunnels started from about DTL3 Ring Segment 499 to 522 (see Fig. 22).

The general tunnelling parameters by DTL3 Drive of Bukit Panjang Bound and Expo Bounds Tunnel are as follows (see Table 7).

Key dates and activities for undercrossing of CCL Outerbound by DTL3 Drive of Bukit Panjang Bound and Expo Bounds Tunnel are as follows (see Table 8).

Under both DTL 3 tunnel drives, the largest vertical movement of CCL structures was recorded at prism installed at shoulder of Outerbound which is the closer bound in the direction of drives. After the first tunnel drive, 6 mm downwards were recorded in which this is corresponded to approximately 0.75% volume loss.

However, during the subsequent Expo Bound tunnelling works, the recorded movements exceeded the predicted vertical movement of 13 mm and eventually stabilised at 17 mm despite that key performance indicator (KPI) remain the same for both bound (see Figs. 23 and 24).

Table 8
Key dates and activities for CCL undercrossing.

Date	Time	Activities
11 Apr 14		A day before TBM1 (BP Bound Drive) entering 1st Reserve Line of CCL Outer bound
13 Apr 14	0500	TBM1 Head (BP Bound Drive) undercrossing CCL Outer bound
14 Apr 14	0500	TBM1 Tail (BP Bound Drive) entering CCL Outer bound
15 Apr 14	2100	TBM1 Tail (BP Bound Drive) out of CCL Outer bound
19 Jul 14		A day before TBM3 (EX Bound Drive) entering 1st Reserve Line of CCL Outer bound
20 Jul 14	1200	TBM3 Head (EX Bound Drive) undercrossing CCL Outer bound
22 Jul 14	0000	TBM3 Tail (EX Bound Drive) entering CCL Outer bound
22 Jul 14	2000	TBM3 Tail (EX Bound Drive) out of CCL Outer bound
25 Jul 14	1500	TBM3 Tail (EX Bound Drive) out of CCL Inner bound 1st Reserve Line

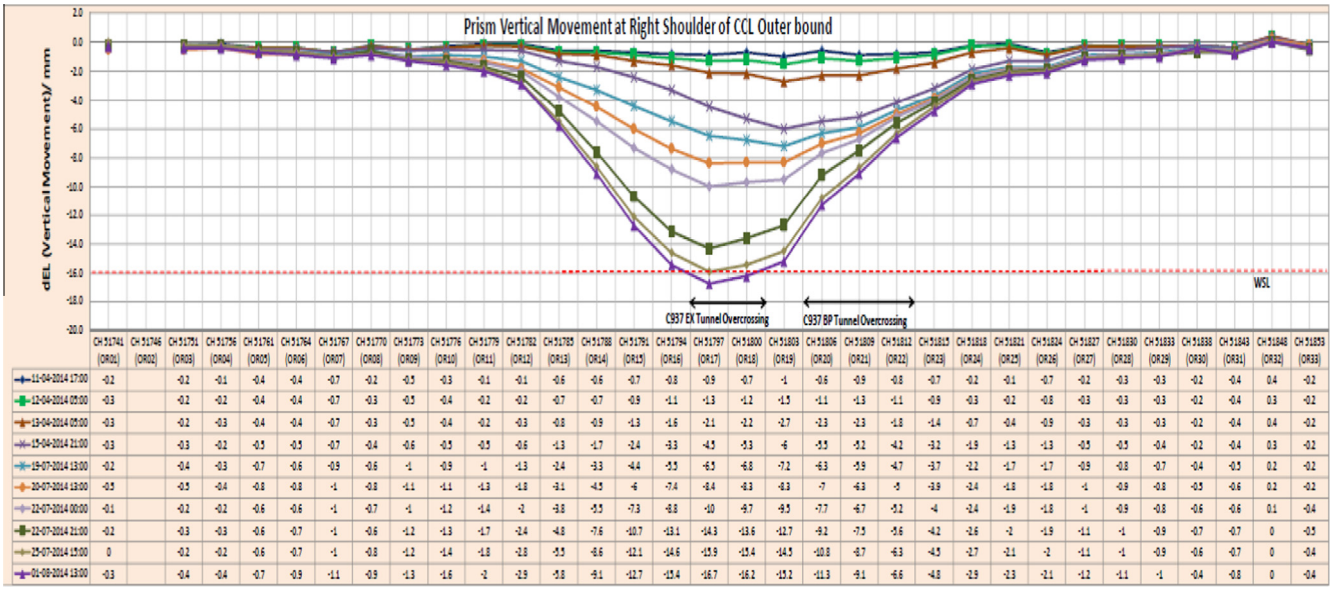


Fig. 23. Recorded vertical movement of CCL live tunnel during undercrossing by DTL3 drives.

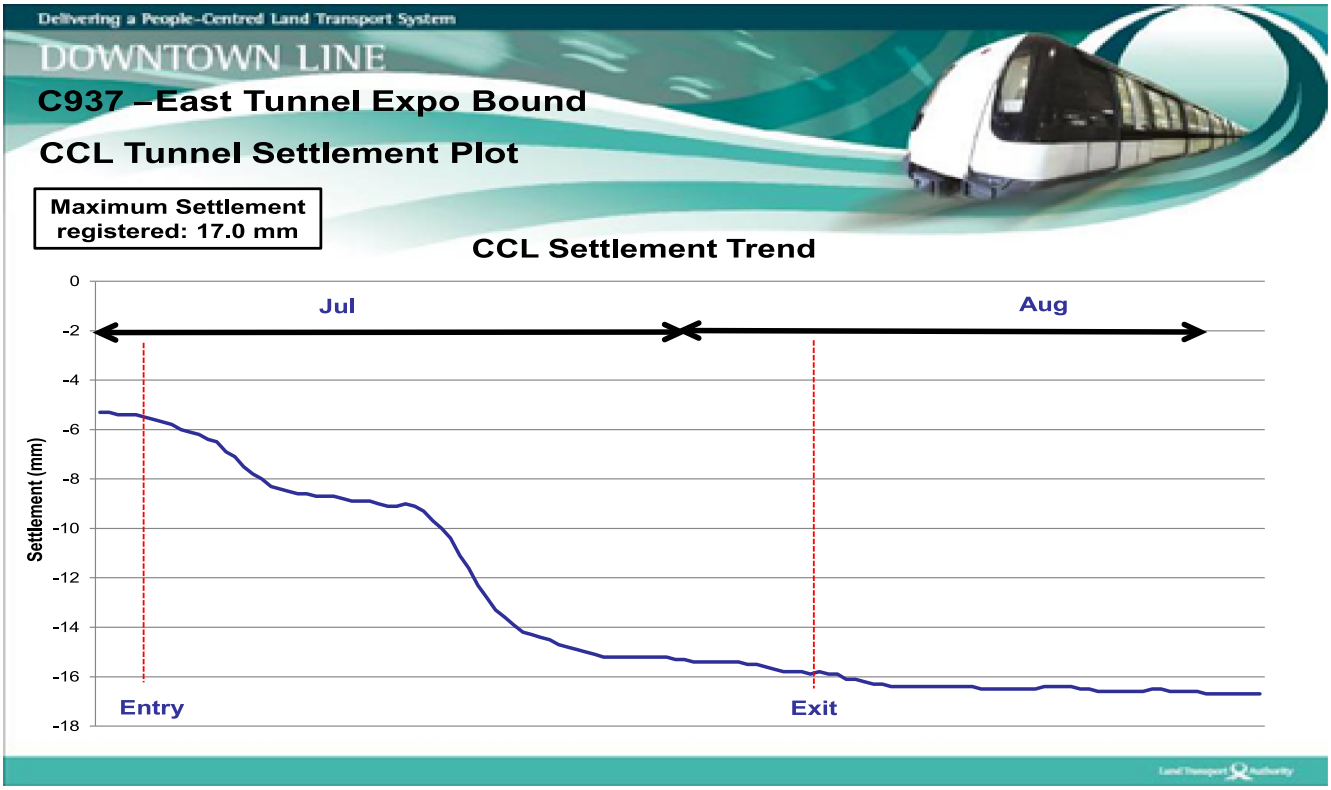
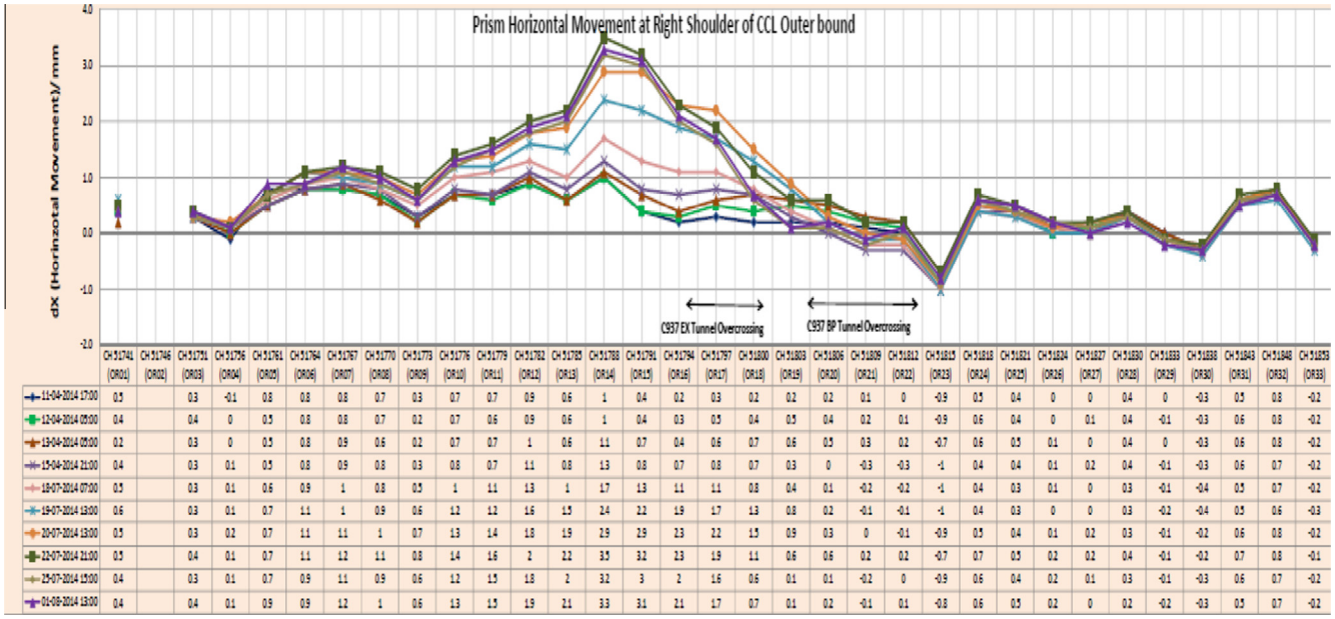
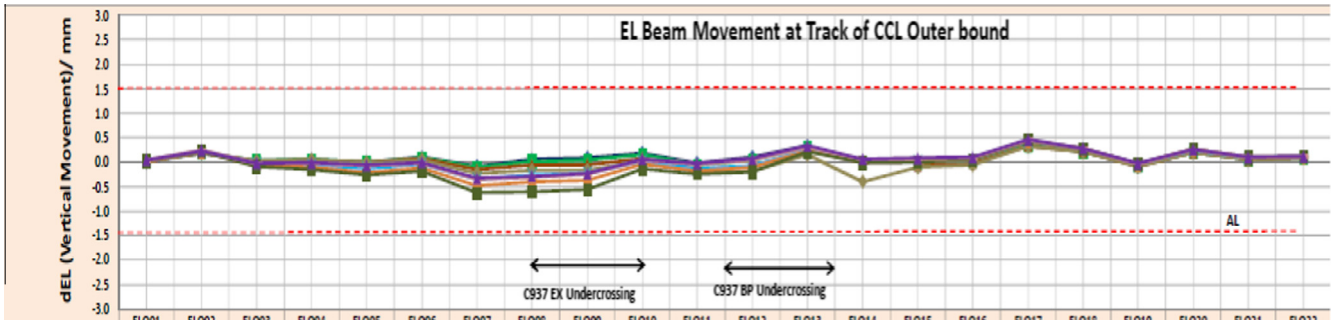


Fig. 24. Settlement Trendplot for second drive TBM.



	CH 51741 (OR01)	CH 51746 (OR02)	CH 51751 (OR03)	CH 51756 (OR04)	CH 51761 (OR05)	CH 51764 (OR06)	CH 51767 (OR07)	CH 51770 (OR08)	CH 51773 (OR09)	CH 51776 (OR10)	CH 51779 (OR11)	CH 51782 (OR12)	CH 51785 (OR13)	CH 51788 (OR14)	CH 51791 (OR15)	CH 51794 (OR16)	CH 51797 (OR17)	CH 51800 (OR18)	CH 51803 (OR19)	CH 51806 (OR20)	CH 51809 (OR21)	CH 51812 (OR22)	CH 51815 (OR23)	CH 51818 (OR24)	CH 51821 (OR25)	CH 51824 (OR26)	CH 51827 (OR27)	CH 51830 (OR28)	CH 51833 (OR29)	CH 51838 (OR30)	CH 51843 (OR31)	CH 51848 (OR32)	CH 51853 (OR33)
11-04-2014 17:00	0.5		0.3	-0.1	0.8	0.8	0.8	0.7	0.3	0.7	0.7	0.9	0.6	1	0.4	0.2	0.3	0.2	0.2	0.2	0.1	0	-0.9	0.5	0.4	0	0	0.4	0	-0.3	0.5	0.8	-0.2
12-04-2014 05:00	0.4		0.4	0	0.5	0.8	0.8	0.7	0.2	0.7	0.6	0.9	0.6	1	0.4	0.3	0.5	0.4	0.5	0.4	0.2	0.1	-0.9	0.6	0.4	0	0.1	0.4	-0.1	-0.3	0.6	0.8	-0.2
13-04-2014 05:00	0.2		0.3	0	0.5	0.8	0.9	0.6	0.2	0.7	0.7	1	0.6	1.1	0.7	0.4	0.6	0.7	0.6	0.5	0.3	0.2	-0.7	0.6	0.5	0.1	0	0.4	0	-0.3	0.6	0.8	-0.2
15-04-2014 21:00	0.4		0.3	0.1	0.5	0.8	0.9	0.8	0.3	0.8	0.7	1.1	0.8	1.3	0.8	0.7	0.8	0.7	0.3	0	-0.3	-0.3	-1	0.4	0.4	0.1	0.2	0.4	-0.1	-0.3	0.6	0.7	-0.2
18-07-2014 07:00	0.5		0.3	0.1	0.6	0.9	1	0.8	0.5	1	1.1	1.3	1	1.7	1.3	1.1	1.1	0.8	0.4	0.1	-0.2	-0.2	-1	0.4	0.3	0.1	0	0.3	-0.1	-0.4	0.5	0.7	-0.2
19-07-2014 13:00	0.6		0.3	0.1	0.7	1.1	1	0.9	0.6	1.2	1.2	1.6	1.5	2.4	2.2	1.9	1.7	1.3	0.8	0.2	-0.1	-0.1	-1	0.4	0.3	0	0	0.3	-0.2	-0.4	0.5	0.6	-0.3
20-07-2014 13:00	0.5		0.3	0.2	0.7	1.1	1.1	1	0.7	1.3	1.4	1.8	1.9	2.9	2.9	2.3	2.2	1.5	0.9	0.3	0	-0.1	-0.9	0.5	0.4	0.1	0.2	0.3	-0.1	-0.2	0.6	0.8	-0.2
22-07-2014 21:00	0.5		0.4	0.1	0.7	1.1	1.2	1.1	0.8	1.4	1.6	2	2.2	3.5	3.2	2.3	1.9	1.1	0.6	0.6	0.2	0.2	-0.7	0.7	0.5	0.2	0.2	0.4	-0.1	-0.2	0.7	0.8	-0.1
25-07-2014 15:00	0.4		0.3	0.1	0.7	0.9	1.1	0.9	0.6	1.2	1.5	1.8	2	3.2	3	2	1.6	0.6	0.1	0.1	-0.2	0	-0.9	0.6	0.4	0.2	0.1	0.3	-0.1	-0.3	0.6	0.7	-0.2
01-08-2014 13:00	0.4		0.4	0.1	0.9	0.9	1.2	1	0.6	1.3	1.5	1.9	2.1	3.3	3.1	2.1	1.7	0.7	0.1	0.2	-0.1	0.1	-0.8	0.6	0.5	0.2	0	0.2	-0.2	-0.3	0.5	0.7	-0.2

Fig. 25. Recorded horizontal movement of CCL live tunnel during undercrossing by DTL3 drives.



	EL001	EL002	EL003	EL004	EL005	EL006	EL007	EL008	EL009	EL010	EL011	EL012	EL013	EL014	EL015	EL016	EL017	EL018	EL019	EL020	EL021	EL022
11-04-2014 17:45	0.0153	0.1885	0.0178	0.0517	-0.0015	0.1021	-0.0817	0.0570	0.0905	0.1892	-0.0166	0.1106	0.3310		-0.0042	0.0226	0.3473	0.2182	-0.0737	0.2027	0.0561	0.0929
12-04-2014 05:46	0.0180	0.1846	0.0180	0.0505	-0.0046	0.0876	-0.1071	0.0138	0.0386	0.1460	-0.0569	0.0432	0.3042		-0.0237	0.0120	0.3354	0.2136	-0.0742	0.2000	0.0533	0.0913
13-04-2014 05:46	0.0185	0.1874	0.0169	0.0473	-0.0100	0.0705	-0.1486	-0.0522	-0.0456	0.0778	-0.1329	-0.0861	0.2431		-0.0680	-0.0173	0.3264	0.2091	-0.0796	0.1976	0.0543	0.0905
15-04-2014 21:49	0.0196	0.1942	0.0232	0.0479	-0.0104	0.0477	-0.2203	-0.1599	-0.1591	0.0192	-0.1868	-0.1750	0.1449	-0.3916	-0.1025	-0.0531	0.3200	0.2076	-0.0846	0.1972	0.0565	0.0910
19-07-2014 13:23	0.0258	0.2184	-0.0388	-0.0525	-0.1253	-0.0448	-0.3482	-0.2569	-0.2241	0.0253	-0.1297	-0.0705	0.2459	-0.0007	0.0110	0.0509	0.4323	0.2677	-0.0866	0.2513	0.0840	0.1114
20-07-2014 12:10	0.0164	0.2115	-0.0734	-0.1059	-0.2087	-0.1198	-0.4760	-0.4024	-0.3730	-0.0375	-0.1737	-0.1056	0.2329	-0.0030	0.0082	0.0503	0.4289	0.2684	-0.0898	0.2521	0.0919	0.1108
22-07-2014 22:51	0.0214	0.2184	-0.0876	-0.1397	-0.2636	-0.1836	-0.6183	-0.6002	-0.5528	-0.1309	-0.2384	-0.1943	0.2112	-0.0085	0.0046	0.0509	0.4385	0.2686	-0.0873	0.2522	0.0975	0.1127
25-07-2014 15:06	0.0411	0.2379	-0.0177	-0.0146	-0.0748	-0.0197	-0.3468	-0.2997	-0.2468	0.0404	-0.0407	0.0306	0.3140	0.0518	0.0713	0.0814	0.4538	0.2753	-0.0835	0.2563	0.1011	0.1131
01-08-2014 13:17	0.0440	0.2359	-0.0184	-0.0057	-0.0582	-0.0046	-0.3254	-0.2838	-0.2301	0.0583	-0.0213	0.0647	0.3303	0.0614	0.0860	0.0965	0.4581	0.2791	-0.0273	0.2605	0.0986	0.1141

Fig. 26. Recorded EL beam movement of CCL track during undercrossing by DTL3 drives.

As for the horizontal movement, larger lateral movement of 3.5 mm was observed at CCL Outerbound during Expo Bound drive (second tunnel drive) even though larger thrust force was applied at early Bukit Panjang Bound drive while applied face pressure remain the same. When Expo Bound drive was approaching CCL Outerbound, tunnel was noticed to be pushed away from the direction of drive. Subsequently, the tunnel seems to be dragged further while TBM is undercrossing and eventually moving towards its initial position when TBM passed through CCL Outerbound (see Fig. 25).

Notwithstanding that recorded vertical movement and differential movement in transverse plane of tunnel structures has exceeded the predicted value, the reading of electro level (EL) beam remain stable and changed insignificantly throughout the entire tunnel drives (see Fig. 26).

7. Conclusion

Overall, the tunnelling over and undercrossing NEL, NSL and CCL tunnels by DTL3 C937 is considered successfully with no sign of structural distress, tensile cracks, opening of circumferential joint and leakage detected in the tunnel structures. The movement of track is also well kept within the safety limit of train operation without having to trigger any reduction in operation train speed.

From the site experience, movement of Live twin MRT tunnel as a result of TBM drive is larger for tunnel which is located nearer to the direction of TBM drive. The existing tunnel, which is located relatively farther to TBM drive, is more resilient to movement due to tunnelling probably because it is shielded by neighbouring tunnel.

It is also noticed that the structure movement of existing live MRT tunnels induced by second tunnel crossing is more than the first drive. In other words, actual combined settlement effect is slightly larger than the movement predicted by FEM 2D model (NSL: actual 11.7 mm vs predicted: 11 mm; CCL: 17 mm vs predicted: 13 mm). As the distance between 2 proposed tunnels are small (less than half of tunnel diameter), the ground could have probably been disturbed at larger extent by the first tunnel drive resulted in a lower geotechnical properties of ground during the second tunnel drive. Hence, the project team recommends a more stringent or restrictive control of tunnelling parameters, excavation volume management and backfill grouting for second tunnel drive in future project.

Besides, it is important to note the percentage of contribution to total settlement due to losses via face pressure, immediate closure of TBM shield, follow with backfill grouting and residual settlement. This will assist to predict the total settlement while crossing of sensitive buildings is still in progress and subsequently to implement improvement measures soonest possible should any undesired movement is registered. From the experience of C937 project, it was observed that 30–40% of total tunnel movement was attributed to losses via Face Pressure & Immediate Closure of TBM Shield, 30 to 50% during backfill grouting stage and 5–10% during residual settlement stage.

Reference

C937 Geotechnical Interpretative Baseline Report, December 2010.